



CS301 LIVE (QUIZ) MCQs

MID TERM

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The process of getting the value of a variable using pointers is called:

- ☐ Dereferencing
- ☐ Memory deallocation
- ☐ Memory allocation
- ☒ Referencing

2. ~BinarySearchTree() is a _____.

- ☐ Switch case
- ☒ Constructor
- ☐ Destructor
- ☐ Template method call

3. Which of the following is TRUE about recursive calls?

- ☐ Recursive calls can only be used with inorder traversal
- ☐ There is no terminating condition in recursive calls
- ☐ Tree data structure cannot use recursive calls
- ☒ Recursive call always contains a terminating condition

4. Symbol is used when we want to get the value of a variable using pointer.



- ☒ *
 - ☐ ||
 - ☐ ::
 - ☐ &
-

5. The _____ symbol is used when we want to get the value of a variable using pointer.

- ☒ *
 - ☐ ||
 - ☐ ::
 - ☐ &
-

6. We can make a lexicographic order of characters based on their _

- ☐ Random choice
 - ☐ Binary digits
 - ☒ ASCII values
 - ☐ Memory addresses
-

7. In which traversal method, the recursive calls can be used to traverse a binary tree?

- ☐ In preorder and inorder traversal only
 - ☒ In preorder, postorder and inorder traversal only
 - ☐ In postorder and preorder traversal only
 - ☐ In postorder and inorder traversal only
-

8. Parameter passing (by value or by reference) is similar to _____.

- ☐ FORTRAN
- ☐ COBOL
- ☒ C++



9. A complete binary tree has 63 non-leaf nodes. The number of leaf nodes will be:

- ☐ 62
- ☐ 64
- ☒ 65
- ☐ 66

10. In the perspective of memory organization every process executing, the second part of the memory is for _____ of the program.

- ☒ Code
- ☐ Static Data
- ☐ Stack

1. If the elements are "3","5",'1',"6", in which order they will be removed from the Queue one by one?

- A) 3156 ✓
- B) 3516
- C) 6513
- D) 6153

2. In the perspective of memory organization, each process is divided into _____ sections.

- A) code, data, and stack ✓
- B) heap, code, and data
- C) data, stack, and registers
- D) stack, heap, and registers

3. In level-order traversal for Binary Search Tree, _____ data structure is used.

- A) Queue ✓



- B) Stack
- C) Heap
- D) Array

4. For making Binary Search Tree for Strings we need,

- A) int
- B) float
- C) double
- D) None of the above ✓

5. The _____ symbol is used when we want to get the value of a variable using a pointer.

- A) ::
- B) ||
- C) *
- D) & ✓

6. Operator have to be destroyed explicitly.

- A) construct
- B) new ✓
- C) build
- D) delete

7. Which of the following is TRUE about recursive calls?

- A) Tree data structure cannot use recursive calls
 - B) There is no terminating condition in recursive calls
 - C) Recursive calls can only be used with inorder traversal
 - D) Recursive call always contains a terminating condition ✓
-



8. Consider an array `int x[] = {14,15,4,9,7,18,3}`. Which statement will be used to set the first number of the list into the root node?

- A) `root->setinfo( &x[0] )` ✓
- B) `setinfo -> root( &x[0] )`
- C) `root->setinfo( [l] )`
- D) `setinfo -> root( &{1} )`

9. What will be the inorder traversal of the given tree?

go

Copy code

```
``` 45 /\ 25 75 /\ 15 35 ```
```

- A) 15 25 35 45 75 ✓
- B) 25 15 35 45 75
- C) 45 25 75 15 35
- D) 15 25 35 75 45

10. If the above condition is satisfied (`this->left == NULL && this->right == NULL`), then the current node is:

- A) Left node
- B) Leaf node ✓
- C) Right node
- D) Root node

1. After deletion of a node from a binary search tree, traversal method should be maintained.

- Level order
- Postorder
- Inorder ✓
- Preorder



2. In which of the following tree for each node, the value of root node is greater than left child and smaller than right child?

- Perfect Binary Tree
- Complete Binary Tree
- **Binary Search Tree** ✓
- Full Binary Tree

3. For searching a particular number in Binary Search Tree (if it is not present), the maximum number of comparisons will be \_\_\_\_\_ comparison(s) at each level.

- **1 comparison** ✓

4. if( this->left == NULL && this->right == NULL ) If the above condition is satisfied then the current node is:

- **Leaf node** ✓
- Root node
- Right node
- Left node

5. ~BinarySearchTree( ) is a \_.

- Template method call
- Switch case
- Constructor
- **Destructor** ✓

6. Two common models of simulation are

- Circuit-based simulation and Object-oriented simulation
- Circuit-based simulation and Time-based simulation
- **Time-based simulation and Event-based simulation** ✓
- Circuit-based simulation and Event-based simulation

7. During in-order traversal using recursive calls, if we found a node is NULL. It means this node will satisfy following condition.

- It will not have both left and right children
- **It will not have left child** ✓



- It will have only left child
- It will not have right child

8. Elements in a queue data structure are added from

- Rear end, front end
- Front end
- Front end, rear end
- Rear end ☒

9. A model attempts to move a rear-world phenomenon

- Simulation
- Logical
- Physical
- Conceptual ☒

10. In time-based simulation, we maintain a/an

- Event
- Register
- Variable ☒

1. If the elements are "3", "5", "1", "6", in which order they will be removed from the Queue one by one?

- 6513
- 3516 ☒
- 3156

---

2. Which of the following calling method does not change the original value of the variable passed to a function as an argument?

- Call by passing as name
- Call by passing as reference
- Call by passing as pointer



- Call by passing as value ✓

---

3. One should be careful about transient \_\_\_\_ that are stored by reference in data structures.

- objects ✓
- function
- stack
- tree

---

4. Which of the following operation returns the contents of a binary tree node?

- isLeaf(p)
- brother(p)
- parent(p)
- info(p) ✓

---

5. For a complete binary tree with n numbers of nodes, the depth is calculated as

- $\log_e (n-1) - 1$
- $\log_g (n-1) + 1$
- $\log_z (n+1) - 1$
- $\log_e (n+1) + 1$  ✓

---

6. `void traversal (TreeNode treeNode) { if( treeNode != NULL ) { traversal (treeNode->getLeft()); traversal (treeNode->getRight()); cout << (treeNode->getinfo())<<" } }`

- Levelorder
- Inorder
- Postorder ✓
- Preorder

---

7. Which one of the following case is the most complicated case to delete a node from BST?



- Node to be deleted is the leaf node
- Node to be deleted has either left child or right child
- No case is complicated
- Node to be deleted has both the left and right children ✓

---

8. In Binary Search Tree, deleting a node is easy if it is a

- Leaf ✓
- Root
- Right
- Left

---

9. Which one is not the property of a binary tree?

- Sibling nodes should have the same parent.
- Every node in binary tree should have maximum two children.
- Non-root nodes should have one parent
- Only one node should have two parents. ✓

---

10. Which one is the correct function call for the following function of calculating the cube?

int cube(int& num) {

- cube(&&num)
- cube(\*num)
- cube(&num)
- cube(num) ✓

1. A complete binary tree has 63 non-leaf nodes. The number of leaf nodes will be:

- ☐ 64
- ☐ 65



- ☒ 62
  - ☐ 63
- 

2. Local variables defined inside function body are:

- ☐ incremented
  - ☐ decremented
  - ☒ destroyed
  - ☐ preserved
- 

3. Left, right, info, and parent are the operations of:

- ☐ Stack
  - ☐ Linked List
  - ☐ Queue
  - ☒ Tree
- 

4. The first statement in the given code is called as:

```
class BinarySearchTree;
```

```
class BinaryNode {
```

```
 friend class BinarySearchTree;
```

```
};
```

- ☐ Post Class Declaration
  - ☐ Binary Declaration
  - ☒ Forward Declaration
  - ☐ Pre Class Declaration
- 

5. From Operating System point of view, the recursive function calls are made with the help of:

- ☒ Stack



- ☐ Queue
  - ☐ Linked list
  - ☐ Binary Search Tree
- 

6. A binary tree is said to be a \_\_\_\_ binary tree if every non-leaf node in a binary tree has non-empty left and right subtrees:

- ☐ AVL
  - ☒ Strictly
  - ☐ Perfect
  - ☐ Complete
- 

7. Two common models of simulation are \_\_\_\_ and \_\_\_\_:

- ☐ Circuit-based simulation and Time-based simulation
  - ☐ Circuit-based simulation and Object-oriented simulation
  - ☒ Time-based simulation and Event-based simulation
  - ☐ Circuit-based simulation and Event-based simulation
- 

8. Level-order traversal for Binary Search Tree can be implemented:

- ☐ Through max-heap
  - ☐ Through both recursive method call and non-recursive method call
  - ☒ Only through non-recursive method
  - ☐ Only through recursive method
- 

9. For searching a particular number in Binary Search Tree (if it is not present), the maximum number of comparisons will be \_\_\_\_ comparison(s) at each level:

- ☒ 1
- ☐ 2



- ☐  $\log(n)$
- ☐  $n$

---

10. In C++, we place the class interface in:

- ☐ .cpp
- ☐ .cppp
- ☐ .hh
- ☒ .h

1. `~BinarySearchTree()` is a

- Switch case
- Destructor
- Template method call
- **Constructor** ☒

2. If we write functions for recursive and non-recursive inorder traversal method of BST, what will be the difference between its functions prototypes?

- Different function names
- **Different arguments list** ☒
- Different return types
- Nothing will be different

3. All the objects created using \_\_\_\_\_ operator have to be destroyed explicitly.

- create
  - construct
  - build
  - **new** ☒
-



4. of the program.

- Stack
- Static Data
- Heap ☒

5. From Operating System point of view, the recursive function calls are made with the help of

- Binary Search Tree
- Queue
- Linked list
- Stack ☒

6. From Operating System point of view, the recursive function calls are made with the help of

- Binary Search Tree
- Queue
- Linked list
- Stack ☒

7. \_\_\_\_\_ is used for Reference variables in C++.

- & ☒
- 
- 
- 
- ::

8. if( this->left == NULL && this->right == NULL )

If the above condition is satisfied then the current node is:

- Root node
- Leaf node ☒



- Right node
- Left node

---

9. \_\_\_\_\_ is an area in computer memory that is used to allocate memory dynamically.

- Stack
- **Heap** ☒
- Static
- Global

---

10. A binary tree is said to be a \_\_\_\_\_ binary tree if every non-leaf node in a binary tree has non-empty left and right subtrees.

- AVL
- Perfect
- **Strictly** ☒

---

11. 

```
void traversal (TreeNode<int> treeNode) { if(treeNode != NULL)
{
traversal (treeNode->getleft());
traversal (treeNode->getRight());
cout << (treeNode->getinfo()) << " "; } }
```

Select the correct option\*

- Preorder
- Inorder
- **Postorder** ☒
- Levelorder

1. In the perspective of memory organization, every process executing, the second part of the memory is for \_\_\_\_\_ of the program.

- Stack ✓
- Static Data



- Code
  - Heap
- 

2. When a node to be deleted from a BST has both left and right child, after deletion its \_\_\_\_\_ will replace this node.

- Inorder successor ✓
  - Predecessor
  - Successor
  - Preorder successor
- 

3. A complete binary tree has 63 non-leaf nodes. The number of leaf nodes will be:

- 65
  - 62
  - 61
  - 64 ✓
- 

4. The process of getting the value of a variable using pointers is called:

- Referencing
  - Memory allocation
  - Dereferencing ✓
  - Memory deallocation
- 

5. A model attempts to model a real-world phenomenon:

- Simulation ✓
  - Conceptual
  - Physical
  - Logical
-



```
6. void traversal (TreeNode treeNode) {
 if(treeNode != NULL) {
 cout << (treeNode->getinfo())<<" traversal (treeNode->getLeft());
 traversal (treeNode->getright());
 }
}
```

What type of traversal is this?

- Preorder ✓
- Inorder
- Postorder
- Sorted order

---

7. One should be careful about transient \_\_\_\_\_ that are stored by reference in data structures.

- function ✓
- stack
- tree

---

8. Which one is the correct function call for the following function of calculating cube?

int cube(int& num)

- cube(num) ✓
- cube(\*num)
- cube(&num)
- cube(&&num)

Each node in singly linked list contains.

- ☐ Two pointers
- ☒ NULL pointer
- ☐ One pointer
- ☐ No pointer

7. Consider the following expression:  $P = A * B + C - D$ . In the above expression "-" is a:

- ☐ Technical



- ☐ Unary
- ☐ Logical
- ☒ Binary

? **isFull() method of stack class will return true when:**

- ☐ Stack is not empty nor full
- ☒ Stack is full
- ☐ Stack is partially full
- ☐ Stack is empty

? **If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?**

- ☒ 5
- ☐ 222

? **Each operator in a postfix expression refers to the previous:**

- ☐ three
- ☐ four
- ☒ one

? **When add() operation of a linked list is called the following action is done:**

- ☒ Set the next pointer of last current node
- ☐ A new node is made
- ☐ Last node is removed to add new
- ☐ Current node is removed to add new

? **In doubly linked list there is/are \_\_\_\_\_ pointer/s in each node.**

- ☐ Four
- ☒ Two
- ☐ One
- ☐ Three

? **Which of the following is the correct conversion of infix to postfix expression?  $2 + B - (D - H) / K$**

- ☐ ZB+DH-K/~
- ☐ ZB+D-H-K/



- ✓ ZB+DHK--/
- O ZB+DH-K-/

2. Which of the following statement is correct for the variable "current--"?

- O current ~ 1 = current
- O current = current ~ 2
- ✓ current = current - 1
- O current = current + 1

2. Convert the given infix form:  $12 + 60 - 23$  of expression in postfix form.

- ✓  $12\ 60 + 23 -$
- O  $12\ 60 + -23$
- O  $+12\ 60 - 23$
- O  $12+ 60\ 23 -$

1. Consider the following expression:  $P = A * B + C - D$ . In the above expression "+" is a

- \_\_\_\_\_
- O Technical
  - O Unary
  - O Logical
  - ✓ Binary

2. Which of the following is not a form of expression?

- O Prefix
- ✓ Pastfix
- O Infix
- O Postfix

3. \_\_\_\_\_ is the major factor to see the efficiency of a program.

- O Code
- ✓ Time
- O Correctness



4. When add() operation of a linked list is called, the following action is done:
- ☐ O Last node is removed to add new
  - ☒ A new node is made
  - ☐ O Set the next pointer of last current node
  - ☐ O Current node is removed to add new
5. Current pointer always points to the \_\_\_\_\_ of the linked list.
- ☐ O Last current node
  - ☒ Last node
6. Trying to remove an element from an empty stack is called:
- ☐ O Empty collection
  - ☐ O Overflow of stack
  - ☐ O Garbage collection
  - ☒ Underflow of stack
7. In simple or singly linked list, there is/are pointer/s in each node.
- ☒ One pointer
8. What will be the postfix expression of the following infix expression?  $D + E * F / G$
- ☐ O  $DE + FG */$
  - ☐ O  $DEF * G / +$
  - ☒  $DE + F * G /$
  - ☐ O  $DE * F / G +$
9. The principal benefit of a linked list over a conventional array is that the order of the linked items may be \_\_\_\_\_ from the order that the data items are stored in memory.
- ☒ Different
  - ☐ O Same
  - ☐ O Equivalent
  - ☐ O Identical
10. Suppose a Stack class has been defined using template. Now, we want to declare a Stack object of an int type. What will be the correct syntax?



- O <int> Stack stack;
- O Stack int stack;
- ✓ Stack<int> stack;
- O int Stack stack;

1. Consider the linked list having data [6, 72, 35, 65, 25] stored in it. While the current pointer is pointing to the memory location having 72 stored in it. After calling the add(4) function on the following linked list, the current pointer will point to the memory location having value?

- A) 25
- B) 35 ✓
- C) 72
- D) 65

2. Consider the linked list having data [6, 72, 35, 65, 25] stored in it. While the current pointer is pointing to the memory location having 35 stored in it. What will be the resultant linked list after calling the add(18) function on the given linked list?

- A) 6, 72, 35, 18, 65, 25 ✓
- B) 6, 18, 72, 35, 65, 25
- C) 6, 72, 35, 65, 25, 18
- D) 6, 72, 18, 35, 65, 25

3. In which data structure are elements inserted at the back and removed from the front?

- A) Linked List
- B) Tree
- C) Queue ✓
- D) Stack

4. Each node in a singly linked list contains two fields, one field called the data field, while the other field contains:

- A) Pointer to character
- B) Pointer to class
- C) Pointer to an integer
- D) Pointer to next node ✓

5. Each node in a singly linked list contains two fields, one field called the data field, while the other field contains:

- A) Pointer to character
- B) Pointer to class
- C) Pointer to an integer
- D) Pointer to next node ✓



---

6. Returns the front element of the queue without removing it:

- A) top()
- B) dequeue()
- C) front() ☒
- D) enqueue(X)

---

7. Trying to remove an element from an empty stack is called:

- A) Garbage collection
- B) Overflow of stack
- C) Underflow of stack ☒
- D) Empty collection

---

8. In the statement template <T>, T represents:

- A) Generic data type ☒
- B) Character data type
- C) Integer data type
- D) String data type

---

9. During the execution of a process, the operating system constructs four things for that process.

Which of the following is not part of that process?

- A) Linked List ☒
- B) Stack
- C) A section for static data including global variables
- D) None of the above

---

10. Consider we have performed the following operations on a stack of size 5: push(10); push(20); push(30); pop(); pop(); push(40); What is the top of the stack?

- A) 40 ☒
- B) 10
- C) 30
- D) 20

---

11. Various cells of memory are not located continuously.

- A) Circular array
- B) Arrays
- C) Linked list ☒

? Each node in a singly linked list contains two fields, one field called data field while other field contains:

- Pointer to next node ☒
- Pointer to class



- Pointer to an integer
- Pointer to character

? Each operator in a postfix expression refers to the previous:

- Three
- One ☒
- Two

? What will be the result of evaluating the following expression?  $5+3*2/(6-3)$ :

- 3
- 7 ☒
- 8
- 10

? New items are added at the \_\_\_ of the stack:

- Center
- Bottom
- Top ☒

?  $2(7+8)-6$  is an example of \_\_\_ expression:\*

- Infix ☒
- Postfix
- Logical

? Consider we have performed the following operations on a stack of size 5. Push(10); Push(20); Push(30); Pop(); Pop(); Push(40); Select the correct option:

- 3 ☒
- 0
- 2

? Consider the linked list having data [6, 72, 35, 65, 25] stored in it. While current pointer is pointing to memory location having 72 stored in it. After calling add(4) function on the following linked list current pointer will point to memory location having value:

- 25 ☒
- 35
- 65
- 72

? A template is a function or class that is written with a \_\_\_:

- Definite
- Generic ☒
- Specific

? The abstract data type refers to the basic mathematical concept that defines the:

- Function type
- Structure
- Class type
- Data type ☒

? In queue data structure elements are removed from:

- Rear



- Push
- Pop
- Front ☒

? Whenever we call a function, the compiler makes a \_\_\_\_\_ that it uses to fulfill this function call.

- Macro
- Queue
- ☒ Heap

? Consider the following push operations of a Stack:

mathematica

Copy code

Stack.push(4);

Stack.push(6);

Stack.push(5);

Stack.push(8);

If a user calls a pop() operation, then which value will be returned?

- 4
- 6
- 5
- ☒ 8

? Consider the following expression:  $P = A * B + C - D$ . In the above expression, "+" is a

- \_\_\_\_\_.
- ☒ Binary
  - Unary
  - Technical

? In singly linked list, a node comprises of \_\_\_\_\_ field/s.

- Three
- 0
- One
- ☒ Two
- Four

? Which one of the following operations returns and removes top value of the stack?

- Push
- First
- Top
- ☒ Pop

? Stack.push(15) will push 15 on:

- Anywhere in the Stack
- ☒ Top of the Stack
- Middle of the Stack
- Bottom of the Stack



❑ Consider the following infix expression:  $7/8+9$ . If one converts the above expression into postfix, what would be the resultant expression?

- $78/+9$
- $178+9$
- $78/9+$
- ☒  $789/+$

❑ In singly linked list, a node consists of two parts:

- ☒ Object and pointer
- Two objects
- Object and structure

❑ At the time while evaluating an expression, if we want to give higher precedence to lower precedence operator, we used \_\_\_\_\_ .

- Comas
- O
- ☒ Parenthesis
- Operands
- Dots

❑ Which of the following signature of the constructor function for a stack is correct as a generic data type?

- `Stack: Stack();`
- `Template Stack: Stack();`
- `Template<stack T>;`
- ☒ `Stack<T> = Stack();`

---

1. In array list the worst case of removing an element is:

- To remove an element from the end of the list
- We cannot remove element from an array list
- To remove an element from the start of the list ☒
- To remove an element from the middle of the list

---

2. Which of the following statement is correct for the variable "current--"?

- `current = current + 1`
- `current = current ~ 2`
- `current = current ~ 1`
- `current = current - 1` ☒

---

3. Consider the linked list implementation of a stack. Which of the following node is considered as Top of the stack?

- Middle node
- First node ☒
- Any node



---

4. is the major factor to see the efficiency of a program.

- Code
- **Quality** ☒
- Correctness

---

5. Each node in singly linked list contains \_\_\_\_\_:

- **One pointer** ☒
- NULL pointer
- No pointer
- Two pointers

---

6. Which of the following is not a data structure?

- Linked list
- Stack
- Queue
- **Memory cell** ☒

---

7. Which of the following is not a form of expression?

- Infix
- Pastfix
- Postfix
- Prefix
- **Pastfix** ☒

---

8. Suppose we have been given the following data set for a Queue: 37524. What will be the resultant Queue if we call a front() method?

- 524
- 7524
- **37524** ☒
- 75243

---

9. A queue is a data structure where elements are:

- inserted and removed from both ends.
- **inserted at the back and removed from the front.** ☒
- inserted at the front and removed from the back.
- inserted and removed from the top.

---

10. An array is collection of cells of the:

- Different type
- Hybrid type
- **Same type** ☒



- Struct type

❑ Which of the following operation is not supported by queue data structure?

- Enqueue(X)
- isEmpty()
- Dequeue()
- **Push() ✓**

❑ In \_\_\_\_\_, a programmer uses two pointers in the node, i.e. one to point to next node and the other to point to the previous node.

- Doubly-linked list ✓
- Array
- Linked list

❑ What will be the postfix expression of following infix expression?  $D + E * F / G$

- $DE * F / G +$
- $DE + FG * /$
- **$DE + F * G /$  ✓**
- $DEF * G / +$

❑ What will be postfix expression of following infix expression?  $4 * 117 + 2$

- $T * + 4172$
- $411 772 +$
- $4177 * 2 +$
- **$* 41772 + **$  ✓**

❑ Consider the linked list implementation of a stack. Which of the following node is considered as Top of the stack?

- Last node
- Middle node
- **First node ✓**

❑ Each operator in a postfix expression refers to the previous

- One
- Three
- Four
- **Two ✓**

❑ Which of the following is the correct conversion of infix to postfix expression?  $2 + B - (D - H) / K$

- $ZB + DH - K - /$
- $ZB + D - H - K /$
- $ZB + DHK - - /$
- **$ZB + DH - K / -$  ✓**

❑ Consider we have performed the following operations on a stack of size 5. Push(10); Push(20); Push(30); Pop(); Pop(); Push(40);

- **10, 40 ✓**

❑ Consider we have performed the following operations on a stack of size 5. Push(10); Push(20); Push(30); Pop(); Pop(); Push(40);



- 0 1 2 3
- **0 1 4 ✓**
  - ☐ Stack can be implemented by which of the following data structure?
- Skip list
- BST
- **AVL**
  - ☐ A stack carries \_\_ behavior.
- UFO
- FIFO
- **LIFO ✓**

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